

Robust Monitoring Machine: A Machine Learning Solution for Out-of-Sample R^2 -Hacking in Return Predictability Monitoring

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Abstract

Out-of-sample R^2 -hacking problems can arise even without multiple testing if a researcher constructs a prediction model using the intuition derived from empirical properties that appear only in the evaluation sample. We provide a machine-learning solution for this problem in the context of robust monitoring where a forecaster switches between a proposed forecast and a benchmark to exploit a dynamic shrinkage effect. In an example of stock return forecasting, the resulting robust monitoring forecast improves the average performance of the proposed forecast by 15% (in terms of mean-squared-error) and reduces the variance of its relative performance by 46% while avoiding the out-of-sample R^2 -hacking problem. Our approach, as a final touch, can further enhance performance and stability of forecasts from any models and methods.

Key words: Machine Learning, Out-of-Sample R^2 -Hacking, Return Predictability, Monitoring

JEL Classification: C52, C53, C55, C58, G17

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1 Introduction

Out-of-sample R^2 is no better than in-sample R^2 when it comes to data snooping concerns. Persistent researchers can attempt multiple prediction models and ingenuously report only the best performing one (Inoue and Kilian 2005, Inoue and Kilian 2006). Or each of careful researchers examines only one model, but only lucky ones obtain false positive results that are good enough for publication (Chordia, Goyal, and Saretto 2017). However, the problem in reality is even deeper than that. Researchers often construct a prediction model using the intuition derived from recent empirical findings that did not exist in the training sample period (Yae 2021). It is unlikely that forecasters back then chose such a model without any hint from the future; i.e., an unintended look-ahead bias arises in pseudo out-of-sample testing.

For example, some of early papers such as Pesaran and Timmermann (1995) show that stock return predictability is time-varying: the predictability is stronger in recessions than expansions. Then many follow-up papers utilize such empirical facts to further improve out-of-sample predictability.¹ The truth is, however, a hypothetical forecaster in out-of-sample prediction simulations would not choose such stylized models without sufficient evidence at the moment of prediction. Therefore, the best option available for the forecaster in the past is only to consider all possible models and choose one from (or combine) them, ex-ante optimally without help of the not-yet-available intuition.

To tackle this comprehensive out-of-sample R^2 -hacking problem, we take a machine learning approach. Our strategy, however, differs from the others that spotlight superior prediction accuracy of machine learning. We instead focus on the downside of machine learning algorithms: their black-box-like nature that blurs economic intuition. Figuratively speaking, using this *black box*, we create a virtual environment that prevents researchers from peeking the intuition from the future when they perform out-of-sample prediction simulations. Therefore, the goal of our approach is not to maximize the average forecasting performance ex-post but to measure the attainable level of out-of-sample predictability in practice.

Our solution for out-of-sample R^2 -hacking is demonstrated in the context of robust monitoring forecasts: a forecast that switches between a proposed forecast and a benchmark while utilizing their conditional predictabilities that are monitored in real-time. To avoid out-of-sample R^2 -hacking, we

¹Among many others, Dangi and Halling (2012) study time-varying coefficient models whose coefficients follow a random walk. Henkel, Martin, and Nardari (2011) and Zhu and Zhu (2013) choose regime switching models instead. Rapach, Strauss, and Zhou (2013) have the discounted mean squared forecast error (DMSFE) in the past determine the combination weights, similarly to Stock and Watson (2004) and Bates and Granger (1969).

suggest the following three-step approach: 1) choose an information set for monitoring, independently of intuition that is available only after the moment of prediction, 2) approximate the entire time-series model space with feature engineering, and 3) apply a combination of ensemble machine learning algorithms, instead of choosing what works best ex-post after attempting many. We design each step to exclude out-of-sample R^2 -hacking associated with selecting data, models, and optimization algorithms, respectively. We call this machine learning approach for robust monitoring forecasts, in short, *robust monitoring machine*.

For demonstration purposes, we apply our robust monitoring machine to a real-world prediction example. The proposed forecast in this example is a combination forecast from [Rapach, Strauss, and Zhou \(2010\)](#) who take an equal-weighted average of the 14 stock market predictors studied in [Goyal and Welch \(2008\)](#). The benchmark forecast is a real-time historical average of stock market returns, as [Campbell and Thompson \(2008\)](#) suggest while defining out-of-sample R^2 . Then we apply the three-step approach as follows. First, we choose the smallest possible information set for monitoring: a history of the prediction loss (e.g., squared-error loss) differences between the proposed and benchmark forecast. Next, following [Christ, Braun, Neuffer, and Kempa-Liehr \(2018\)](#), we convert the variable in our information set into hundreds of time-series features, which act as building blocks to approximate the entire time-series model space. Finally, we simply combine three popular ensemble machine learning algorithms: Random Forest, Extremely Randomized Trees, and Gradient Boosting.

Out-of-sample classification performance of robust monitoring machine is still outstanding despite our efforts to avoid out-of-sample R^2 -hacking problems. Its sensitivity is 57.0% and specificity is 56.8%, which statistically differ from those of uninformative random classifiers. Our results imply that the past return-predictability truly contains information for the future return predictability of proposed and benchmark forecasts.

To quantify the advantages of robust monitoring machine, we measure a prediction loss difference (between a forecast and the benchmark forecast). Its mean represents the average forecasting performance while its variance does the stability of forecasting performance. We find that our monitoring forecast beats the proposed forecast in both aspects. The mean of prediction loss difference improves by 15% because of our machine learning approach while the variance falls by 46.3% because of robust monitoring's expected benefits. Inputs for our approach are nothing but the forecasts made by other methods. Therefore, robust monitoring machine can be easily added to other forecasting methods as a final step rather than competes with them. In particular, fore-

casters should apply robust monitoring machine if their performances are evaluated relative to the benchmark.

We also notice that robust monitoring machine captures the well-known economic intuition in real-time: return predictability concentrates in bad economic times. Robust monitoring machine tends to favor the proposed forecast ex-ante in bad economic times which are characterized by low stock valuation and high macro uncertainty. Previously, this intuition looks clear only when researchers look at the whole sample period, as a hindsight.

This paper contributes to two strands of the literature. First, a few recent papers use a forward-looking approach to model selection or averaging; [Zhu and Timmermann \(2017\)](#), [Gibbs and Vasnev \(2018\)](#), and [Granziera and Sekhposyan \(2019\)](#) optimize weighting strategy conditional on expected future performance of the prediction models; this is a big departure from the previous literature with the backward-looking approaches. Then we fully scale up their forward-looking approach by searching the entire time-series model space that is approximated by feature engineering, instead of considering only a few models possibly chosen by subjective intuitions. Second, stock return prediction by machine learning shows a tremendous progress recently both in academia and practice. Prevailing researches, however, focus on the improved accuracy of the prediction.² Instead, we exploit the black-box-like nature of machine learning so that we can intentionally block intuition from the hypothetical future data (evaluation sample) in out-of-sample simulations. To the best of our knowledge, this paper is the first attempt to use machine learning in order to resolve the out-of-sample R^2 -hacking problem rather than create one.

The remainder of the paper is organized as follows. Section 2 revisits forecast monitoring in a simple framework. Section 3 explains the benefits of robust monitoring forecasts. Section 4 outlines our machine learning approach to robust monitoring. We describe its real-world example in Section 5 and present the results in Section 6. We discuss how to interpret the results in conjunction with business cycles in Section 7 and conclude in Section 8.

2 Monitoring Forecasts Revisited

Monitoring forecasts can be viewed as an extreme case of combination forecasts with dynamic weights. If a researcher is unsure which forecast predicts conditionally best, it is better to combine multiple forecasts for variance reduction. Yet, selecting one single predictor can be better if an

²See, for example, [Gu, Kelly, and Xiu \(2020\)](#), [Feng, Giglio, and Xiu \(2020\)](#), [Heaton, Polson, and Witte \(2017\)](#), and [Freyberger, Neuhierl, and Weber \(2020\)](#) among many others.

accurate signal for conditional performance is available. In this section we characterize the condition on which a monitoring forecast can outperform individual forecasts in a simple model. We also include a few metrics and statistical tests to quantify monitoring performance.

2.1 A Simple Model of Monitoring

Suppose a target variable to predict is r and its unconditional mean is μ . There are two unbiased forecasts $f^{(a)}$ and $f^{(b)}$, i.e., $E[f^{(a)}] = E[f^{(b)}] = E[r] = \mu$, as follows.

$$r = \mu + e^{(y)} + e^{(s)}, \quad (1)$$

$$f^{(a)} = \mu + e^{(a)} + S e^{(s)}, \quad (2)$$

$$f^{(b)} = \mu + e^{(b)} + (1 - S) e^{(s)}. \quad (3)$$

The individual shocks $(e^{(y)}, e^{(s)}, e^{(a)}, e^{(b)})$ follow a normal distribution centered at zero, respectively. These shocks are uncorrelated to each other, and their standard deviations are $SD(e^{(y)}) = \sigma_y$, $SD(e^{(s)}) = \sigma_s$, and $SD(e^{(a)}) = SD(e^{(b)}) = \sigma_f$, respectively. The indicator variable S follows a Bernoulli distribution with probability p_s .

$$S = \begin{cases} 1 & \text{with probability } p_s, \\ 0 & \text{with probability } 1 - p_s. \end{cases} \quad (4)$$

Without loss of generality, we assume that the forecast $f^{(a)}$ is unconditionally more accurate than $f^{(b)}$ such that $p_s > \frac{1}{2}$. If $S = 1$, the forecast $f^{(a)}$ is conditionally more accurate than $f^{(b)}$ ex-ante while the forecast $f^{(b)}$ is so if $S = 0$.

Now consider a monitoring forecast $f^{(m)}$ optimally switching between two forecasts $f^{(a)}$ and $f^{(b)}$ given a signal on S . The accuracy of the signal is measured by p_m as follows.

$$f^{(m)} = M f^{(a)} + (1 - M) f^{(b)} \quad \text{where} \quad M = \begin{cases} S & \text{with probability } p_m, \\ 1 - S & \text{with probability } 1 - p_m. \end{cases} \quad (5)$$

M is a monitoring signal following a Bernoulli distribution. The higher p_m is, the more accurate monitoring forecast is. For example, if $p_m = 1$, the monitoring forecast always picks the more

accurate forecast between $f^{(a)}$ and $f^{(b)}$. Next, we adopt a squared forecast error as a loss function,

$$L^{(a)} = (r - f^{(a)})^2, \quad L^{(b)} = (r - f^{(b)})^2, \quad \text{and} \quad L^{(m)} = (r - f^{(m)})^2, \quad (6)$$

and define a loss difference between two forecasts as follows.

$$\Delta L^{(b,a)} = L^{(b)} - L^{(a)}, \quad \Delta L^{(b,m)} = L^{(b)} - L^{(m)}, \quad \text{and} \quad \Delta L^{(a,m)} = L^{(a)} - L^{(m)}. \quad (7)$$

Then we can easily derive the condition the monitoring forecast $f^{(m)}$ outperforms both individual forecasts on average: The monitoring forecast $f^{(m)}$ is unconditionally more accurate than $f^{(a)}$ (and so $f^{(b)}$) ex-ante if and only if

$$E[\Delta L^{(a,m)}] = E[\Delta L^{(b,m)}] - E[\Delta L^{(b,a)}] = (p_m - p_s)\sigma_s^2 > 0. \quad (8)$$

The condition (8) says that the monitoring forecast will outperform if a monitoring signal is accurate enough to dominate the accuracy advantage of an individual forecast over the other. For example, in case two forecast are similar in accuracy $p_s = 1/2$, then it is not difficult to beat both forecasts by monitoring: we need only $p_m > 1/2$.

2.2 Out-of-Sample R^2 : A Conventional Metric for Forecast Performance

Campbell and Thompson (2008) suggests an out-of-sample R^2 metric to evaluate the performances of stock return forecasts. Suppose $f_{t|t-1}^{(i)}$ is a given forecast for the target r_t . Then the out-of-sample R^2 for the period from $t = t_0$ to $t = t_1$ is given by

$$R_{OS}^2 = 1 - \frac{\sum_{t=t_0}^{t_1} (r_t - f_{t|t-1}^{(i)})^2}{\sum_{t=t_0}^{t_1} (r_t - f_{t|t-1}^{(b)})^2}. \quad (9)$$

where $f_{t|t-1}^{(b)}$ is a benchmark forecast which is the historical average of the past returns r_t , commonly used in the stock return predictability literature. The out-of-sample R^2 measures the reduction in mean square prediction error for a forecast relative to that for the benchmark forecast. If R_{OS}^2 is positive, the forecast $f_{t|t-1}^{(i)}$ outperforms the benchmark forecast in terms of the mean square prediction error metric.

2.3 Metrics and Tests for Monitoring Performance

Monitoring forecasts solves a classification problem; which forecast performs better conditionally? Therefore, we can adopt metrics and statistical tests from the classification literature to evaluate monitoring performance. For example, we can define the event that $f^{(a)}$ outperforms $f^{(b)}$, i.e., $\Delta L^{(b,a)} > 0$, is *positive* whereas *negative* otherwise. Then *sensitivity*, or true positive rate (TPR), refers to the empirical probability that the monitoring forecast $f^{(m)}$ equals $f^{(a)}$ when $\Delta L^{(b,a)} > 0$ (i.e., $f^{(a)}$ outperforms $f^{(b)}$). Likewise, *specificity*, or true negative rate (TNR), refers to the empirical probability that $f^{(m)}$ equals $f^{(b)}$ when $\Delta L^{(b,a)} < 0$. We also adopt other metrics such as positive predicted value (PPV), negative predictive value (NPV), and accuracy (ACC) following their conventional definitions in a confusion matrix.

If the classifier is purely random, both ‘sensitivity + specificity’ and ‘PPV + NPV’ should be one in population. We compute their 95% confident intervals as follows.

$$TPR + TNR \pm 1.96 \times \sqrt{TPR \times (1 - TPR)/n_1 + TNR \times (1 - TNR)/n_2}$$

where n_1 and n_2 are the numbers of true positives and negatives in the data, respectively.

$$PPV + NPV \pm 1.96 \times \sqrt{PPV \times (1 - PPV)/n_3 + NPV \times (1 - NPV)/n_4}$$

where n_3 and n_4 are the numbers of predicted positives and negatives in the data, respectively. If these confidence intervals do not contain one, we can conclude that a monitoring task is informative ($p_m > 1/2$) at 5% significance level.

We adopt two formal test methods for monitoring performance from the classification context: Fisher’s Exact test and Chi-square test for binary classifiers. The null hypothesis H_0 in both test methods is that true (predicted) positives and true (predicted) negatives are equally likely to be predicted as (true) positives. Therefore, low p-values of these tests are evidence that a monitoring task is informative.

3 Robust Monitoring

Monitoring Forecast is considered as an aggressive technique to maximize predictability in contrast to *Combination Forecast* which aims to reduce variance of forecast errors. That is, monitoring forecast and combination forecast are traditional counterparts of boosting and bagging (i.e., bootstrap

aggregating) in machine learning, respectively. However, monitoring forecast, which is aggressive by nature, can produce a conservative predictor with robustness when it switches between a given proposed forecast and a benchmark. That is, monitoring forecast becomes a robustified version of the originally proposed forecast, whatever it is. We briefly explain the intuition in the following sections.

3.1 Using a Benchmark to Robustify a Forecast

Suppose a researcher wants to measure forecasting ability of a proposed forecast $f^{(a)}$ relative to a benchmark forecast $f^{(b)}$ in the context of the previous section. Then the loss difference, $\Delta L^{(b,a)} = L^{(b)} - L^{(a)}$, between the forecasts $f^{(a)}$ and $f^{(b)}$ is a measure of how much the forecast $f^{(a)}$ outperforms the benchmark forecast $f^{(b)}$. With a quadratic loss function, the expected loss difference $E(\Delta L^{(b,a)})$ is the expected reduction in mean squared error when we replace the benchmark $f^{(b)}$ by $f^{(a)}$.

On the other hand, a researcher can construct a monitoring forecast $f^{(m)}$ that switches between a proposed forecast $f^{(a)}$ and a benchmark forecast $f^{(b)}$. She wonders if the monitoring forecast $f^{(m)}$ outperforms the proposed forecast $f^{(a)}$. Note comparing the mean squared errors of two forecasts $f^{(a)}$ and $f^{(m)}$ is equivalent to comparing the expected loss differences $E(\Delta L^{(b,m)})$ and $E(\Delta L^{(b,a)})$ because of the following identity:

$$E[L^{(a)}] - E[L^{(m)}] = E[\Delta L^{(b,m)}] - E[\Delta L^{(b,a)}].$$

Here, the empirical loss difference ΔL is a main building block for relative forecasting performance evaluation. Its first moment is a difference in the mean squared errors which is a key comparison metric in forecasting literature. A researcher, therefore, compares their means $E(\Delta L^{(b,m)})$ and $E(\Delta L^{(b,a)})$ to see if the monitoring forecast $f^{(m)}$ outperforms the original forecast $f^{(a)}$ on average. However, what about their variances $Var(\Delta L^{(b,m)})$ and $Var(\Delta L^{(b,a)})$? Should a researcher ignore or care about them? What do they even mean?

Those variances represent the uncertainty of how well the forecasts $f^{(a)}$ and $f^{(m)}$ perform, at a given time, relative to the benchmark forecast $f^{(b)}$, respectively. Yae (2018) adopts the idea of TEV-efficient (tracking-error-volatility efficient) frontier from the portfolio optimization literature and argues that a risk-averse researcher should care about those variances if her performance is evaluated relative to the benchmark forecast.³ The idea of relative performance is commonly used

³Roll (1992) defines tracking error volatility as a square root of the sample second moment of differences in a portfolio and benchmark return. The TEV-efficient frontier shows a trade-off between relative risk premium and

in investments. An investor who wants to outperform the market would consider deviating from the market as taking risk for higher returns. The more her portfolio deviates from the market, the greater downside risk (and upside opportunity) she takes relative to the market performance.

Nevertheless, conventional forecasting performance metrics focus only on the mean performances while ignoring information in the variance of performances. Such second moment information is used only in some formal tests of relative forecast performance such as [Diebold and Mariano \(2002\)](#).⁴ This first-moment-oriented practice should look alarming to financial economists since it ignores risk which is the core of investment performance evaluations. Considering the variance of relative performance is actually common in economics and finance.⁵ For example, ‘Keeping-up-with-the-Joneses’ preferences says that an agent’s utility is determined relative to others’ consumption level. Similarly, stock market movements do not compensate or penalize a mutual fund manager whose official benchmark is the market portfolio.

Robust monitoring forecast is defined as ‘a monitoring forecast switching between a proposed forecast and a benchmark’. It has a built-in shrinkage effect when the forecast is evaluated relative to the benchmark in terms of $\Delta L^{(b,m)}$. The monitoring technique in this context robustifies a new proposed forecast as well as exploiting shifts in the conditional predictability between two forecasts. Its mechanism is simple. If monitoring is informative, the monitoring forecast will choose to be a benchmark forecast $f^{(m)} = f^{(b)}$ when the other forecast $f^{(a)}$ is unlikely to outperform the benchmark. Whenever $f^{(m)} = f^{(b)}$, the loss difference $\Delta L^{(b,m)}$ becomes exactly zero, and the total variance of the loss difference turns lower than that of the proposed forecast $f^{(a)}$. In the monitoring model of the previous section, the law of total variance implies

$$\begin{aligned} Var(\Delta L^{(b,m)}) &= E[Var(\Delta L^{(b,m)}|M)] + Var[E(\Delta L^{(b,m)}|M)] \\ &= p_a Var(\Delta L^{(b,a)}) + p_a(1 - p_a)\{E(\Delta L^{(b,a)})\}^2, \end{aligned}$$

where $p_a \equiv Prob[f^{(m)} = f^{(a)}] = p_m p_s + (1 - p_m)(1 - p_s)$ denotes the unconditional probability that the monitoring forecast deviates from the benchmark, $f^{(m)} = f^{(a)}$. The second term is negligible

relative risk to the benchmark while the standard efficient frontier in the portfolio theory is a special case where the benchmark is the risk-free asset.

⁴Note this second moment is not the variance of forecast errors but that of differences in squared forecast errors, defined relative to the choice of benchmark. That is, the variance of relative performance is related to the fourth, not the second, moment of forecast errors.

⁵However, unlike the first moment, the second (central) moment comparison requires caution with existence of the benchmark forecast:

$$Var(L^{(a)}) - Var(L^{(m)}) \neq Var(\Delta L^{(b,a)}) - Var(\Delta L^{(b,m)}).$$

unless the proposed forecast $f^{(a)}$ outperforms the benchmark significantly and persistently, which implies the benchmark is improperly chosen as a straw man. Therefore, the ratio of variance of loss differences $\Delta L^{(b,m)}$ can be approximated as p_a :

$$\frac{Var(\Delta L^{(b,m)})}{Var(\Delta L^{(b,a)})} \approx p_a \leq p_s < 1 \quad \text{and} \quad p_a \in [1 - p_s, p_s]. \quad (10)$$

This variance ratio in the left-hand-side is always lower than one by construction. Therefore, monitoring with a benchmark forecast always lowers the variance of loss difference $\Delta L^{(b,m)}$, and the monitoring forecast will produce more statistically significant evidence on superior forecasting performance. This is a hidden benefit of forecast monitoring.

However, such a benefit is never a free lunch. There is a trade-off between accuracy gain and variance reduction in loss difference as follows.⁶

$$\underbrace{\left[\frac{2p_s - 1}{\sigma_s^2} \right]}_{\text{Constant (+)}} \cdot \underbrace{\left[E(\Delta L^{(b,m)}) - E(\Delta L^{(b,a)}) \right]}_{\text{Increase in average accuracy}} + \underbrace{(1 - p_a)}_{\substack{\text{Reduction} \\ \text{in variance} \\ \text{of loss diff.}}} = \underbrace{2p_s(1 - p_s)}_{\text{Constant (+)}}. \quad (11)$$

To derive this, we eliminate p_m by combining $p_a = p_m p_s + (1 - p_m)(1 - p_s)$ and Equation (8). Informative monitoring (i.e., high p_m) improves the accuracy of the monitoring forecast but decreases the reduction in loss-difference variance as follows.

$$\frac{\partial(1 - p_a)}{\partial p_m} = 1 - 2p_s < 0 \quad \text{if} \quad p_s > \frac{1}{2}. \quad (12)$$

Note this trade-off is beyond the bias-variance trade-off in statistical estimators or machine learning algorithms. The bias-variance trade-off is to maximize the average forecasting performance. Instead, Equation (11) represents a trade-off between the average forecasting performance and its variance. This new kind of trade is about the mean and variance of squared forecast errors which correspond to the second and fourth moments of forecast errors while the bias-variance trade-off is about the first and second moments of forecast errors. In a numerical example with $p_s = p_m = 3/4$, we can still expect 37.5% reduction in variance of loss-difference with the same accuracy as the originally proposed forecast in terms of the mean squared error.

The definition of variance of loss difference depends on choice of the benchmark forecast. In many applications, the benchmark is not a subjective matter. For example, the classical view on

⁶This trade-off relationship is from [Yae \(2018\)](#).

the stock market is that the stock price follows a random walk and there is no predictability at all. In other words, the market risk premium is unconditionally constant and investors do not have any conditioning variables to predict it. If a researcher wants to show existence of a successful predictor or predictability of market risk premium, she needs to set up the constant market risk premium as a null hypothesis and use a historical average of stock market returns as a benchmark forecast.⁷

3.2 Metrics for Robust Monitoring Performance

We adopt two metrics for robust monitoring performance from Yae (2018). The metrics are designed as an analogy to investment performance measures: risk premium as a raw metric and alpha as a risk-adjusted metric.⁸ Here, we use concise notations for performance metric inputs:

$$d_m \triangleq \Delta L^{(b,m)} \quad \text{and} \quad d_a \triangleq \Delta L^{(b,a)}.$$

Monitoring Risk Premium Suppose a researcher chooses a forecast $f^{(a)}$ while considering a benchmark forecast $f^{(b)}$ as a reference point. Then $E[d_a]$ can represent her average forecasting performance as the expected reduction in mean squared forecast errors relative to the benchmark. If a researcher chooses $f^{(b)}$ instead, she gets only $E[d_b]$ which is zero by definition. We interpret $E[d_a]$ as a kind of premium in the forecasting context. As an analogy, if an investor choose the stock market portfolio over her benchmark risk-free asset, the expected return difference between these two is called a market risk premium. The investor deviates from the benchmark risk-free asset in hope of earning the premium. Likewise, a researcher deviates from the benchmark forecast $f^{(b)}$ to $f^{(a)}$ in hope of earning $E[d_a]$. She can earn $E[d_m]$ instead by switching between $f^{(a)}$ and $f^{(b)}$ as a robust monitoring forecast $f^{(m)}$. Then $E[d_m]$ scaled by $E[d_a]$ is called *monitoring risk premium*.

$$(\text{Monitoring Risk Premium}): \quad RP_m = \frac{E[d_m]}{E[d_a]} = \frac{R_{OS,m}^2}{R_{OS,a}^2}. \quad (13)$$

where $R_{OS,a}^2$ and $R_{OS,m}^2$ are the out-of-sample R^2 for the proposed forecast $f^{(a)}$ and the robust monitoring forecast $f^{(m)}$ that switches between $f^{(a)}$ and the benchmark forecast $f^{(b)}$, respectively. If monitoring is sufficiently informative as the condition (8), RP_m can be larger than one.

⁷Section 5 shows an empirical example.

⁸We do not consider a utility-function-based measure such as a certainty equivalent although, for example, smooth ambiguity preference (Klibanoff, Marinacci, and Mukerji 2005) can internalize $\text{Var}(\Delta L^{(b,m)})$ and $\text{Var}(\Delta L^{(b,a)})$.

Monitoring Alpha Monitoring risk premium measures average forecasting performance but ignores any risk adjustment. By contrast, imagine investors who use CAPM alpha (or alpha from multi-factor models) as a risk-adjusted investment performance metric. Similarly, the monitoring alpha is defined as follows:

$$(\text{Monitoring Alpha}): \alpha_m = \frac{E[d_m] - E[d_{m^*}]}{E[d_a]}. \quad (14)$$

where m^* is the uninformed monitoring forecast that shifts randomly between the forecasts $f^{(a)}$ and $f^{(b)}$. It is a random strategy whose mixing probability is set so that its $E[d_{m^*}^2]$ equals $E[d_m^2]$. Thus $E[d_{m^*}]$ represents accuracy gain by randomly shifting between the forecasts, and the real gain by informative monitoring should not include $E[d_{m^*}]$. For example, if $\alpha_m = 0.4$, informative monitoring adds 40% of relative accuracy of the proposed forecast on top of the benefit from random switching. Alternatively, monitoring alpha can be also expressed as follows.⁹

$$\alpha_m = \frac{E[d_m]}{E[d_a]} - \frac{E[d_m^2]}{E[d_a^2]}. \quad (15)$$

The first term is a scaled total accuracy gain by monitoring, i.e., monitoring risk premium, while the second term adjusts the relative risk increased by monitoring procedure. When the unpredictable component is large in its scale σ_y , we have $(E[d_m])^2 \ll \text{Var}[d_m]$, so $E[d_m^2] \approx \text{Var}[d_m]$. Therefore, the monitoring alpha is analogous to a utility level of mean-variance preference. Note monitoring alpha of two forecasts $f^{(a)}$ and $f^{(b)}$ are zero by definition.

4 Robust Monitoring Machine

Data snooping is a common issue in empirical research. The problem arises when a researcher reports only the best model or statistically significant variable after numerous failed trials. The same problem can also take place, however, when numerous researchers try only one model or variable, but only a few of researchers can successfully publish their studies depending on their luck as shown in [Chordia, Goyal, and Saretto \(2017\)](#). Such a fundamental problem of empirical research is difficult to avoid, and it persists even in the forecasting context. Robust monitoring is not an exception.

⁹It is easy to show $E[d_{m^*}]/E[d_{m^*}^2]$ is invariant and so $E[d_{m^*}]$ is linearly proportional to $E[d_{m^*}^2]$.

4.1 Out-of-Sample R^2 -Hacking Problem

Data snooping (or p-hacking) comes from two root causes: data and models (algorithms). First, a researcher faces infinite combinations of choices on variables, sample periods, and training-evaluation sample splits. Second, she also need to choose a model along with tuning parameters, algorithms, and estimation methods. Analyzing the whole data-model space exceeds an individual researcher's cognitive ability. Therefore, she ends up selecting one or a few combinations arbitrarily or deliberately which might even worsen data snooping issues.

The p-hacking problem regarding variable choices is well known in the in-sample fitting context, but the same problem also exists in out-of-sample analysis regarding both variable and model choices. For example, in the return predictability literature, out-of-sample R^2 is mainly used as a forecasting performance metric. Nonetheless, out-of-sample R^2 (or any other cross-validation metric) is still exposed to the multiple testing problem and look-ahead bias. For example, some of early papers such as [Pesaran and Timmermann \(1995\)](#) show that predictability is time-varying and conditional on other variables. That is, the predictability is stronger in recessions than expansions. Then many follow-up papers internalize such empirical facts in a specific time-series model to further improve out-of-sample predictability, ex-post.

The pitfall of this practice is that researchers select such successful models and variables based on their intuition that did not exist in the sample period of the back tests. The intuition is formed by empirical knowledge only available now but unavailable at the beginning of the out-of-sample test period, such as information that the predictability is stronger in recession than expansions. Therefore, applying such models and variables since the beginning of the out-of-sample test period is highly unlikely for the real forecasters at that time due to lack of evidence back then. That is, many sophisticated models inspired by such ex-post intuition are all subject to this unintended look-ahead bias. The best option for a forecaster back then, if feasible, is to compare all possible models and variables to find the best one (or best combination) ex-ante since the beginning of the out-of-sample testing period while repeating the process sequentially.

4.2 A Machine Learning Solution

Machine learning algorithms are often criticized because of their black-box-like nature. Despite their superior prediction ability, they make researchers blind to hidden mechanisms by blurring economic intuition. Here we focus on the bright side of the black-box-like nature and transform

such criticism into a crucial device in our study. We make our solution for the robust monitoring problem intentionally blind to any intuition based on the information in the evaluation sample period, as Yae (2021) suggests. Therefore, the goal of our approach is to confirm the existence of useful information for monitoring in the real-time data rather than maximize the average forecasting performance ex-post. To achieve this goal, we implement the following three steps.

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1. Choose a set of conditioning variables for monitoring, independent of information from the evaluation sample period.
2. Approximate the entire time-series model space with feature engineering.
3. Apply a combination of ensemble machine learning algorithms, instead of choosing what works best ex-post after trying many.

We call this machine learning approach *robust monitoring machine* and implement its each step as follows. First, we make the most parsimonious choice on conditioning variables rather than the most universal. In monitoring, a researcher needs to have at least one forecast target and two competing forecasts. We use a single time-series loss difference between two competing individual forecasts, $\Delta L_t^{(b,a)}$, to predict its next sign. The other extreme is to consider the entire information space, yet it is difficult to fathom, collect, or even approximate. Furthermore, lots of data sources are private or highly costly and so out of reach from researchers. Since the entire information space is neither known nor accessible, random sampling from it is also impossible. Second, we expand a single time-series of a conditioning variable $\Delta L_t^{(b,a)}$ into hundreds of time-series features as building blocks to approximate the entire time-series model space. Finally, we combine multiple ensemble machine learning algorithms. We rely on machine learning algorithms to handle the complexity of the feature set, but avoid cherry-picking an algorithm that appears to work best ex-post.

5 Applications: Robust Monitoring for Return-Predictability

We apply our robust monitoring machine to the US stock market prediction problem. Section 5.1 explains the data source and our choice on the benchmark and proposed forecasts. Section 5.2 describes our three-step approach in detail.

5.1 Data, Benchmark, and Proposed Forecasts

The random-walk hypothesis is a traditional view on the stock markets. If investors are rational and fully utilize public information to price stocks, such public information should not predict future stock returns. Following the literature in return predictability, we set the target variable to predict, as monthly stock market excess returns: continuously compounded returns $r_{sp,t+1}$ on the S&P 500 index, including dividends, in excess of the risk-free rate $r_{f,t}$ implied by the Treasury bill rate.¹⁰ Henceforth we call the target variable simply ‘return’.

$$\text{Target Variable: } r_{t+1} = r_{sp,t+1} - r_{f,t}. \quad (16)$$

Suppose the random-walk hypothesis is true. Then econometricians will find that no publicly available variables are correlated with the subsequent return r_{t+1} or its conditional expectation $E[r_{t+1}|\mathcal{F}_t]$ where $\mathcal{F}_t$ denotes the information set of econometricians at time t . The expected return, therefore, should look like a constant to econometricians $E[r_{t+1}|\mathcal{F}_t] = E[r_{t+1}]$ whether it truly is or not. The natural benchmark forecast under the random-walk hypothesis as a null will be an estimate for the unconditional expected return $E[r_{t+1}]$. That is, the historical average of the past returns is the benchmark forecast.

$$\text{Benchmark Forecast: } f_{t+1|t}^{(b)} = \frac{1}{t} \sum_{\tau=1}^t r_{\tau}. \quad (17)$$

The idea of the random-walk hypothesis is theoretically appealing and relates to the well-known efficient market hypothesis. Most of all, it was difficult to reject the hypothesis anyway in early days due to an insufficient sample size. Later researchers, however, began to find some statistical evidence on a few variables predicting stock market returns based on in-sample regression analysis.¹¹ However, [Goyal and Welch \(2008\)](#) examine the out-of-sample performance of the real-time individual OLS regression estimators of 14 variables $x_{i,t}$ and find none of them have out-of-

¹⁰Since the risk-free rate is known at the time of forecast, predicting raw returns is informationally equivalent to predicting excess returns.

¹¹It is worth noting that the rejection of the random-walk hypothesis does not necessarily mean the stock market is inefficient in information processing or investors are irrationally inattentive. It simply means the expected return is time-varying and correlated with some publicly available variables because, roughly speaking, investors’ risk tolerance is time-varying. For example, in recession investors become less risk-tolerant because of their reduced income and wealth. They avoid investing in stocks even if they know the expected return is higher than in the boom periods.

sample predictability. From the following 14 individual predictive regressions:

$$r_{t+1} = \alpha_i + \beta_i x_{i,t} + \epsilon_{i,t+1} \quad \text{for } i = 1, \dots, 14. \quad (18)$$

Goyal and Welch (2008) define the real-time individual forecasts as follows.

$$f_{t+1|t}^{(i)} = \hat{\alpha}_{i,t} + \hat{\beta}_{i,t} x_{i,t}, \quad (19)$$

where $\hat{\alpha}_{i,t}$ and $\hat{\beta}_{i,t}$ are the OLS coefficient estimates of α_i and β_i , respectively, using information up to time t . Later, following Hendry and Clements (2004) and Timmermann (2006) and many others, Rapach, Strauss, and Zhou (2010) construct the equal-weighted average of these 14 individual forecasts as follows and show it outperforms the benchmark forecast (17) in out-of-sample prediction.

$$\text{Proposed Forecast: } f_{t+1|t}^{(a)} = \frac{1}{14} \sum_{i=1}^{14} f_{t+1|t}^{(i)} \quad (20)$$

We use this equal-weight combination forecast as our proposed forecast in monitoring.¹² Then we will optimally switch our choice on forecast between the benchmark forecast in (17) and this proposed forecast in (20).

The monthly data for returns and predictor variables are obtained from Amit Goyal's website.¹³ The sample period starts from January 1927 to December 2017. The 14 predictor variables in our example include dividend-price ratio (dp), dividend yield (dy), earning-price ratio (ep), dividendpayout ratio (de), equity risk premium volatility (rvol), book-to-market ratio (bm), net equity expansion (ntis), treasury bill rate (tbl), long-term yield (lty), long-term return (ltr), term spread (tms), default yield spread (dfy), default return spread (dfr), and inflation (infl). See Appendix A for detailed descriptions.

5.2 Robust Monitoring Machine in Action

This section explains in details how we implement robust monitoring machine.

¹²Alternatively, we can optimize the combination weights or construct a new forecast utilizing nonlinearity and interactions between 14 variables. However, our goal is to build a robust and general approach that can be applied to forecasts from human forecasters or different models/algorithms as well.

¹³The data used in this paper can be found at <http://www.hec.unil.ch/agoyal/> along with detailed descriptions.

Labeling for Binary Classification We define a new dependent variable y_t as follows to convert the monitoring problem to a binary classification.

$$y_t = \begin{cases} 1, & \text{if } \Delta L_t^{(b,a)} > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

where $\Delta L_t^{(b,a)} = (r_t - f_{t|t-1}^{(b)})^2 - (r_t - f_{t|t-1}^{(a)})^2$ is the difference in squared forecast errors as a loss function. Target variable r_t , benchmark forecast $f_{t|t-1}^{(b)}$, and proposed forecast $f_{t|t-1}^{(a)}$ are as defined in Section 5.1. This binary time-series variable y_t , as a label, indicates which forecast outperform the other, ex-post. Note the resulting dependent variable y_t is identical whether the definition of $\Delta L_t^{(b,a)}$ is based on L^2 -norm or L^1 -norm.

Features for Monitoring As we discuss in Section 4.2, we do not introduce any arbitrary conditioning variables helping a monitoring task. Instead, we use a single time-series of $\Delta L_{t-1}^{(b,a)}$ to predict y_t . Following Christ, Braun, Neuffer, and Kempa-Liehr (2018), we perform feature engineering on it. We transform the past 60-month of $\Delta L_t^{(b,a)}$ into 441 time-series model features Z_{t-1} .¹⁴ These features include entire characteristics of the time-series such as the number of peaks, the average, maximal value, autocorrelation, and linear trend.¹⁵

$$g : X_{t-1} \rightarrow Z_{t-1} \text{ where } X_{t-1} = \{\Delta L_{t-1}^{(b,a)}, \Delta L_{t-2}^{(b,a)}, \dots, \Delta L_{t-60}^{(b,a)}\}.$$

The features for monitoring Z_{t-1} offer building blocks to approximate the entire space of time-series models that predict y_t .

Three Ensemble Decision-Tree Algorithms To handle a number of features, we use the three flagship ensemble decision-tree algorithms to predict y_t : Random Forest, Extremely Randomized Trees, and Gradient Boosting. They can effectively accommodate nonlinearity and interaction in features: They avoid overfitting problems in traditional logistic regressions by combining the forecasts from many small trees into a single forecast. We summarize technical differences of these three algorithms in Appendix B.

¹⁴Using the past sixty month is a standard practice in finance literature due to changing nature of the market: for example, CAPM beta estimation.

¹⁵See TSFRESH package in Python.

Training and Tuning Parameters To predict y_t , we train the tree using input-label pairs $(y_{t-\tau}, Z_{t-\tau-1})$ for $\tau = 1, 2, \dots, 120$ (ten years of monthly data). We choose a tuning parameter that maximizes the ROC-AUC statistic measure which is the AUC (Area Under The Curve) of ROC (Receiver Operating Characteristics) curve. Roughly speaking, this procedure maximizes true positive rates relative to false positive rates.¹⁶ The out-of-sample forecast starts from January 1947 following Goyal and Welch (2008).

Ensemble At time t , three different algorithms will produces their best guesses on the probability of $y_{t+1} = 1$, say, $Prob[y_{t+1} = 1|A_i, \mathcal{F}_t]$ for $i = 1, 2, 3$ where A_i is an algorithm and $\mathcal{F}_t$ is an information set. Then we compute the equal-weight average of such probabilities relying on the wisdom of crowds in the algorithm domain.

$$Prob[y_{t+1} = 1|\mathcal{F}_t] = \frac{1}{3} \sum_i^3 Prob[y_{t+1} = 1|A_i, \mathcal{F}_t], \quad (22)$$

Our monitoring forecast on y_{t+1} at time t will be determined by the following rule.

$$f_{t+1|t}^{(m)} = \begin{cases} f_{t+1|t}^{(a)}, & \text{if } Prob[y_{t+1} = 1|\mathcal{F}_t] > \frac{1}{2}, \\ f_{t+1|t}^{(b)}, & \text{otherwise.} \end{cases} \quad (23)$$

6 Empirical Performance of Monitoring Forecasts

6.1 Performance of the Proposed Forecast

The first row of Table 1 shows the performance of the proposed forecast $f^{(a)}$ in Equation (20). The out-of-sample R^2 of the proposed forecast is 0.5% when the evaluation sample starts from 1947. The performance metric, however, is unstable and sensitive to the sample split date. When we gradually change the date from 1947 to 2007, the out-of-sample R^2 decreases and even becomes negative since 1987. The proposed forecast performs poorly showing no predictability ($R_{OS}^2 = -0.24\%$) when the evaluation sample starts from 2007 and ends in 2017.

Figure 1 Panel (a) also confirms that the performance of the proposed forecast $f^{(a)}$ significantly

¹⁶We split these 120 sample pairs into three of 40 sample pairs D_1 , D_2 , and D_3 chronologically. Then we train the tree using D_1 and test on D_2 . Again, we train the tree using D_1 and D_2 and test on D_3 . We choose the optimal tuning parameters based on these two test results.

varies over time with a downward trend. The time-series in the plot is the 60-month trailing moving average of the loss difference between the benchmark and the combination forecasts $\Delta L_t^{b,a}$, which measures the performance of the proposed forecast relative to the benchmark. The plot shows two deep negative values implying that the proposed forecast often greatly underperforms the benchmark, especially, in recent periods. Such an unstable and deteriorating performance can be a serious concern to the investors although the proposed forecast overall outperforms the benchmark during the testing period 1947-2007.

Instability of the proposed forecast's performance is a fundamental problem of a class of combination forecasts suggested by Rapach, Strauss, and Zhou (2010). We explore different weighting schemes in combination. First, we use the median of 14 forecasts instead of their mean. Second, following Stock and Watson (2004) Equation (4), we compute the combination weights of 14 individual forecasts based on their DMSFE (Discounted Mean Squared Forecast Errors) with two tuning parameters: 1) the trailing sample period and 2) discount factor δ . The instable pattern of performance, however, still exist in all cases of different combination weights.

For example, out-of-sample R^2 can rise up to 1.17% with weights proportional to DMSFE (up to the past one month data with a discount factor $\delta = 1$). Yet the out-of-sample R^2 drops down to -1.26% when the evaluation sample starts from 2007 and ends in 2017. Here, suppose we focus only on the average performance metric $R_{OS}^2=1.17\%$ and argue that we find a superior forecast. If we really do so, this is cherry-picking, an example of out-of-sample R^2 -hacking. Furthermore, it comes with a high price tag. Such a cherry-picked forecast performs the worst among all in the recent decade. This instability of performance is a red flag that implies cherry-picking practices.

6.2 Robust Monitoring Performance

We first evaluate how well our robust monitoring machine can conditionally choose between $f^{(a)}$ and $f^{(b)}$. A confusion matrix in Table 2 analyzes the monitoring performance as a classification problem. Sensitivity and specificity are 57.0% and 56.8%, respectively, both higher than 50%. The 95% analytic confidence interval of the sum of sensitivity and specificity is (1.07, 1.21), above one, implying informative monitoring. Note the sum of sensitivity and specificity should be one if a classifier is uninformative and purely random. Similarly, PPV (Positive Predicted Value) and NPV (Negative Predicted Value) are 55.5% and 58.3%, both greater than 50%. The 95% analytic confidence interval of the sum of PPV and NPV is (1.07, 1.21), also above one. Furthermore, the p-values of Fisher's exact test and Chi-square test are both less than 0.01%. Our robust monitoring

machine is informative in the out-of-sample prediction context.

Figure 2 visualizes two performance metrics for robust monitoring. The monitoring risk premium RP_m is a ratio of $E[d_m]$ to $E[d_a]$. The metric RP_m is larger than one since $E[d_m] > E[d_a]$, and therefore the monitoring forecast $f^{(m)}$ outperforms the proposed forecast $f^{(a)}$ in term of average out-of-sample R^2 . On the other hand, the monitoring alpha α_m is a ratio of $E[d_m] - E[d_{m*}]$ to $E[d_a]$, and this metric adjusts the risk in forecasts with respect to the benchmark forecast $f^{(b)}$. Note $E[d_{m*}]$ represents the baseline performance of an uninformative monitoring forecast. Thus, the monitoring alpha is the net gain of performance by our robust monitoring machine, in addition to the risk reduction by robust monitoring described in Equation (10) and (11). The monitoring premium and alpha of our robust monitoring machine is 1.15 and 0.61, respectively, as shown in Table 3. Their Bootstrap p-values are lower than 5%. Robust monitoring machine truly predicts which forecast performs better. Our monitoring performance metrics are consistent with Diebold-Marino (DM) test results. The p-values of a DM test for robust monitoring machine is 0.4% while 6.8% for the proposed forecast. Robust monitoring machine shows stronger statistical significance (lower p-values) since $E[d_m] > E[d_a]$ but $Var[d_m] < Var[d_a]$. The monitoring forecast boosts the average loss difference by 15.1% but reduces its variance by 46.3%, which is a typical benefit of robust monitoring.

Figure 1 Panel (b) repeats Panel (a) but for robust monitoring machine, showing 60-month trailing average of $\Delta L_t^{(b,m)}$. This forecasting performance metric still fluctuates but rarely drops below zero. Its variation is much lower compared to the proposed forecast in Figure Panel (a). Table 4 shows out-of-sample R^2 with different sample split dates. Unlike the proposed forecast $f^{(a)}$, the out-of-sample R^2 of the robust monitoring forecast $f^{(m)}$ never become negative. For the full sample period starting 1947, the out-of-sample R^2 is 0.57% for the robust monitoring forecast while 0.50% for the proposed forecast. Our robust monitoring machine increases 14% in average performance in terms of out-of-sample R^2 while reducing the variation of conditional performance. A shrinkage forecast which is a simple average of $f^{(a)}$ and $f^{(b)}$ fails to improve the performance. We include a few alternative approaches to robust monitoring for comparison. They all fail to improve the proposed forecast; that is, the superior performance of robust monitoring machine is not easy to achieve.

6.3 Certainty Equivalent Return

Following [Campbell and Thompson \(2008\)](#) and [Ferreira and Santa-Clara \(2011\)](#), we compute the certainty equivalent return (CER, henceforth) for an investor with a mean-variance preference who monthly allocates her capital across equities and risk-free bills using market return forecasts. At the end of month t , the investor optimally allocates the share w_t of the portfolio to a market index fund and the remaining share $1 - w_t$ to a risk-free bill. The investor holds the position until the end of month $t + 1$ and repeats her asset allocation task every month. Then the optimal share w_t can be computed as

$$w_t = \frac{1}{\gamma} \frac{E[r_{t+1}|\mathcal{F}_t]}{Var[r_{t+1}|\mathcal{F}_t]}, \quad (24)$$

where r_{t+1} is the excess return (raw return less risk-free rate) and $E[r_{t+1}|\mathcal{F}_t]$ and $Var[r_{t+1}|\mathcal{F}_t]$ are its conditional mean and variance. Following [Campbell and Thompson \(2008\)](#), we assume the followings. First, the investor replaces the conditional mean and variance by a given return forecast and simple variance estimator from the past five-year monthly returns. Second, we restrict the share proportion w_t to lie between 0 and 1.5. Third, we assume the relative risk coefficient γ to be 5. Then the CER measure of the portfolio is defined as

$$CER = \hat{\mu}_p - \frac{1}{2}\gamma\hat{\sigma}_p^2, \quad (25)$$

where $\hat{\mu}_p$ and $\hat{\sigma}_p^2$ are the time-series average and variance estimate, respectively, of the investor's portfolio return $r_{p,t+1}$ over the forecast evaluation period. The portfolio return $r_{p,t+1}$ for month $t + 1$ can be easily calculated as $r_{p,t+1} = w_t r_{t+1} + (1 - w_t) r_{f,t+1}$, ex-post. Unlike out-of-sample R^2 , CER measure explicitly accounts for the risk taken by an investor during the out-of-sample test period. The CER can be interpreted as the risk-free rate of return that an investor is willing to trade with her optimal risky portfolio.

The CER gain is the difference between the CER for the investor who uses any candidate forecast $f^{(i)}$ of the market return and the CER for an investor who uses the historical average benchmark forecast $f^{(b)}$.

$$(\text{CER gain of forecast } f^{(i)}) = CER^{(i)} - CER^{(b)}. \quad (26)$$

We multiply CER gain by 1,200 so that it represents the annual percentage portfolio management fee that an investor would be willing to pay to have access to the given forecast $f^{(i)}$ instead of the

historical average benchmark forecast $f^{(b)}$.

Table 5 repeats Table 4 but reports the annualized certainty equivalent return gains: the economic values of forecasting market returns by given forecasts instead of the benchmark forecast. We observe the same pattern in the outcomes confirming the results expressed in terms of out-of-sample R^2 in Table 4. The CER gain is 1.05% from 1947 to 2017 while the combination forecast has CER gain of 0.90%. The improvement gets larger when we evaluate more recent samples since 2007 (0.84 over 0.40) and since 1997 (1.22 over 0.36). Therefore, CER gains by robust monitoring machine is relatively stable over time.

7 Discussion: Empirical Relationship to the Economy

Mounting evidence shows that return predictability concentrates in bad economic times.¹⁷ Such an established empirical fact produces hindsight theories that try to explain it and allows prediction models to take advantage of it for better ex-post performance.¹⁸ However, the real question to ask is if we can create a prediction model that can foresee the concentration of return predictability before the observed data reveals it. To answer this question, we compare our robust monitoring machine outputs with two variables that represent bad economic times.

We first consider a variable for stock valuation. The log dividend-price ratio of S&P500 is commonly used in the literature as valuation ratio and also a return predictor. High dividend-price ratios mean low valuation of stocks and so bad economic times. Figure 3 displays both log dividend-price ratio (dotted line) and the computed real-time probability p_t (solid line) that the proposed forecast $f^{(a)}$ will outperform the benchmark $f^{(b)}$, calculated by robust monitoring machine. The fluctuation patterns of two graphs are similar. When stock valuation is low, robust monitoring machine favors the proposed forecast over the benchmark. Figure 4 plots the macro uncertainty measure from Jurado, Ludvigson, and Ng (2015) (dotted line) and the probability p_t (solid line). When macro uncertainty is high (usually high at bad economic times), robust monitoring machine favors the proposed forecast over the benchmark. From these two plots, we confirm that our robust monitoring machine can foresee the concentration of return predictability before the observed data explicitly reveals it to econometricians.

¹⁷See Rapach, Strauss, and Zhou (2010), Henkel, Martin, and Nardari (2011), and Dangel and Halling (2012) among others.

¹⁸Cujean and Hasler (2017) provide theoretical model to explain why stock return predictability concentrates in bad times.

8 Conclusion

We provide a machine-learning solution for the out-of-sample R^2 -hacking problem in robust monitoring. The resulting forecast improves the average performance of a proposed forecast by 15% and reduces its variance of performance by 46.3%. Diebold-Mariano test statistic becomes significant, with its p-value falling from 0.068 to 0.004. The sensitivity and specificity of monitoring as a classifier are 57.0% and 56.8%, respectively, which are statistically different from those from random classifiers. Robust monitoring machine predicts time-variation of return-predictability over business cycles without a look-ahead bias.

The proposed forecast in our application is a combination forecast from [Rapach, Strauss, and Zhou \(2010\)](#), yet we can apply robust monitoring machine to any forecast to improve and robustify its performance. Therefore, professional forecasters can use our approach as a final touch to any sophisticated prediction models of their choices. Our approach only helps other forecasting methods rather than competes with them. Also, our three-step approach can be implemented in many different ways for further practical improvements.

The underlying idea of our approach can extend to other fields in finance. For example, we can construct a trading strategy for a mutual fund manager whose performance is evaluated based on the benchmark portfolio. Using machine learning techniques, a fund manager can deviate from the benchmark only when the signal is strong enough to earn extra trading profits. We leave such extensions for future research.

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Figure 1: Time-Varying Performances of the Proposed and the Monitoring Forecasts

This figure plots the performances of the proposed and the monitoring forecasts relative to the benchmark. Positive values mean a forecast outperforms the benchmark. The horizontal axis is year while the vertical axis is the 60-month trailing moving average of difference in squared forecast error, multiplied by 10^4 .

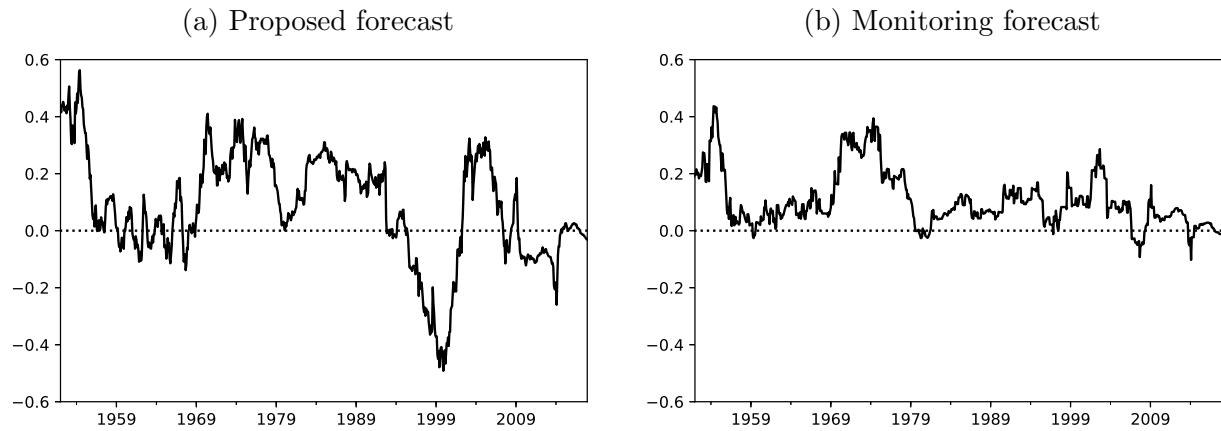


Figure 2: Risk and Returns in Forecasting Performance

This figure visualizes forecasts in the plot of risk and returns. The vertical axis is the the expected loss difference that represents the average forecasting performance relative to the benchmark. The horizontal axis is the expected squared loss difference as risk in forecasting performance. The benchmark forecast, by definition, is located at the origin. The proposed forecast $f^{(a)}$ and the monitoring forecast $f^{(m)}$ are also located in the plot according to their performance and risk metrics.

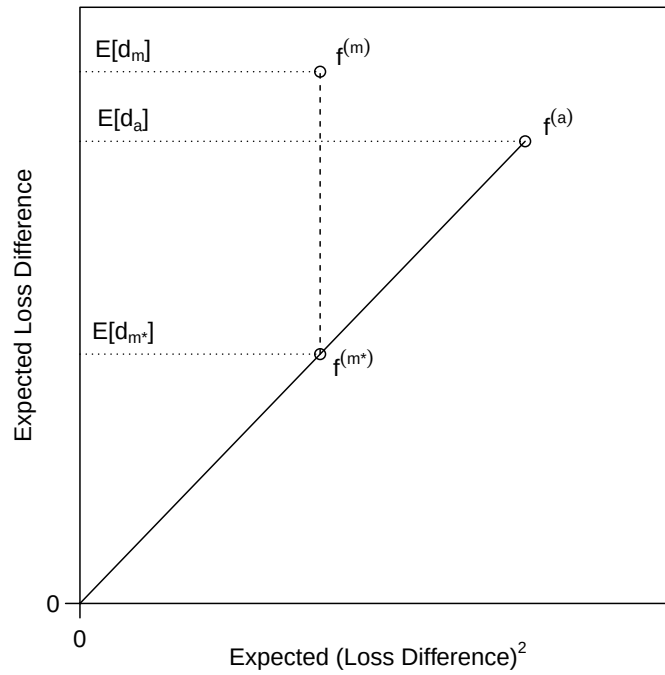


Figure 3: Return Predictability and Stock Valuations

This figure shows the log dividend-price ratio (dotted line) and the probability that the proposed forecast outperforms the benchmark, computed by robust monitoring machine (solid line). The probability in the figure is shown as its 24-month trailing moving-average. The horizontal axis is year.

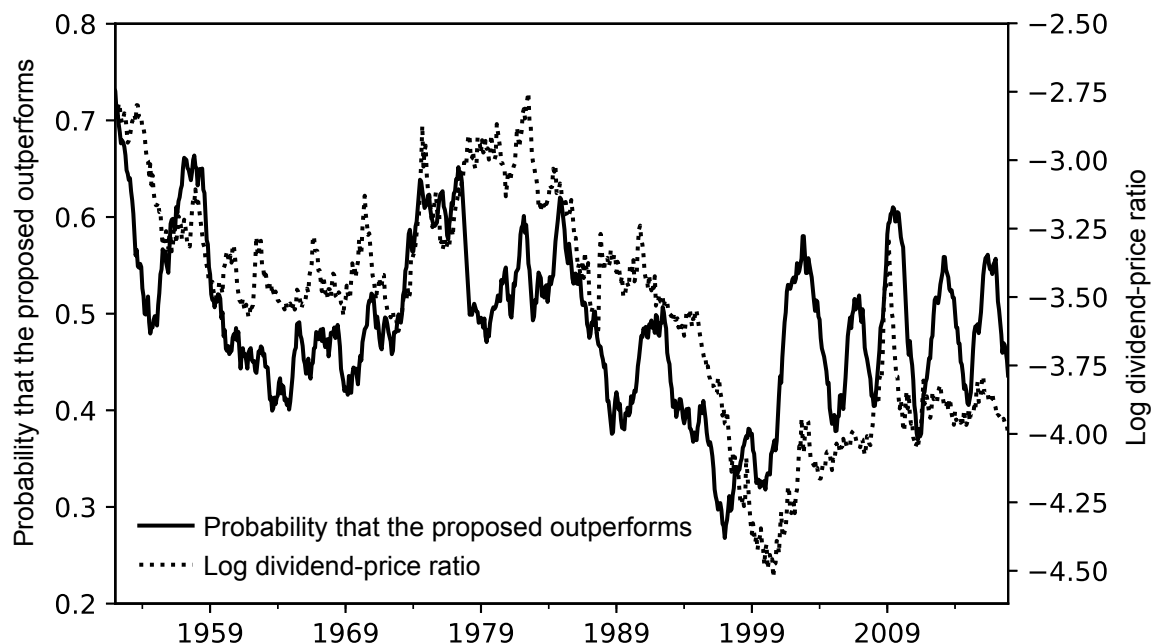


Figure 4: Return Predictability and Macroeconomic Uncertainty

This figure shows the macroeconomic uncertainty (dotted line) and the probability that the proposed forecast outperforms the benchmark, computed by robust monitoring machine (solid line). The probability in the figure is shown as its 24-month trailing moving-average. The horizontal axis is year.

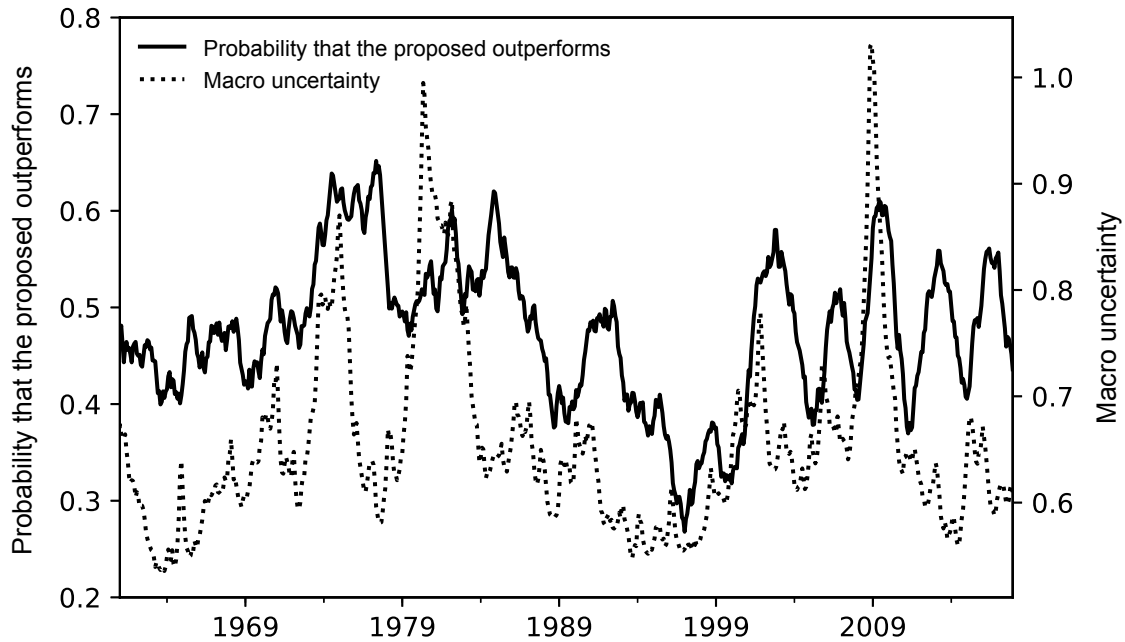


Table 1: Out-of-sample R^2 (%) of Traditional Combination Forecasts

This table reports the forecasting performance of the proposed forecast and its variations, in terms of out-of-sample R^2 (%). The main proposed forecast is a equal-weighted combination forecast from [Rapach, Strauss, and Zhou \(2010\)](#). Its variational forms include a combination as a median and combinations using discounted mean square forecast error (DMSFE) from [Stock and Watson \(2004\)](#). We explore different holdout windows of 1, 12, 24 and 60 months and discount factors of 0.5 and 1. Each column corresponds to a different sample split year from 1947 to 2007. All evaluation period ends in December 2017.

	First year in evaluation sample						
	1947	1957	1967	1977	1987	1997	2007
Proposed forecast: $f^{(a)}$	0.50	0.37	0.36	0.14	-0.09	-0.10	-0.24
Alternative combination forecasts							
Median	0.40	0.37	0.38	0.21	0.09	0.08	0.04
DMSFE (60 months, $\delta = 1.0$)	0.50	0.37	0.37	0.15	-0.08	-0.09	-0.24
DMSFE (24 months, $\delta = 1.0$)	0.49	0.36	0.37	0.14	-0.04	-0.03	-0.19
DMSFE (12 months, $\delta = 1.0$)	0.56	0.43	0.42	0.18	-0.03	-0.00	-0.14
DMSFE (1 month, $\delta = 1.0$)	1.17	1.09	1.18	1.13	-0.31	-0.34	-1.26
DMSFE (60 months, $\delta = 0.5$)	0.57	0.45	0.43	0.14	-0.08	-0.01	-0.08
DMSFE (24 months, $\delta = 0.5$)	0.57	0.45	0.43	0.14	-0.08	-0.01	-0.08
DMSFE (12 months, $\delta = 0.5$)	0.57	0.45	0.43	0.14	-0.08	-0.01	-0.08

Table 2: Monitoring Performance Measures

This table evaluates the performance of out-of-sample binary classification by robust monitoring machine for the evaluation period from 01/1947 to 01/2017. Panel A shows a confusion matrix based on predicted binary outcomes: positive (the proposed forecast outperforms the benchmark) or negative (otherwise). Sensitivity, or True Positive Rate (TPR), is calculated as the number of true positives divided by the number of real positives. Specificity, or True Negative Rate (TNR), is computed as the number of true negatives over the number of real negatives. PPV stands for Positive Predicted Value, calculated as the number of true positives divided by the number of predicted positives. NPV, or Negative Predictive Value, is computed as the number of true negatives divided by the number of predicted negatives. Accuracy is equal to the number of correctly predicted as a fraction of the total number of the sample observations. Panel B reports the 95% confident intervals for ‘Sensitivity + Specificity’ and ‘PPV + NPV’.

Panel A: Confusion Matrix			
	True Positive	True Negative	
Predicted Positive	236	189	PPV = 55.5%
Predicted Negative	178	249	NPV = 58.3%
	TPR = 57.0% (Sensitivity)	TNR = 56.8% (Specificity)	Accuracy = 57.0%
Panel B: Classification Performance Tests			
	Estimate	95% C.I.	P-value
TPR + TNR	1.14	(1.07, 1.21)	
PPV + NPV	1.14	(1.07, 1.21)	
Fisher’s Exact Test			6.84×10^{-5}
Chi-square Test			5.29×10^{-5}

Table 3: Robust Monitoring Performance Measures

This table reports monitoring performance measures: monitoring risk premium and monitoring alpha. $E[d.]$ and $Var[d.]$ denote the expected value and variance of loss difference, which represent average forecasting performance and its relative risk, respectively. Diebold-Mariano tests are to show if a forecast has significantly different predictability from a benchmark. The (two-sided) p-values in parentheses are calculated via bootstrapping.

Forecast		Monitoring Risk Premium	Monitoring Alpha	$E[d.]$ $\times 10^{-6}$	$Var[d.]$ $\times 10^{-8}$	Diebold-Mariano Statistic
Robust Monitoring	estimate (p-value)	1.15 (0.017)	0.61 (0.021)	9.81	0.99	2.87 (0.004)
Proposed	estimate (p-value)	1	0	8.52	1.86	1.82 (0.068)
Benchmark	estimate	0	0	0	0	N/A

Table 4: Out-of-sample R^2 (%) of Monitoring Forecasts

This table repeats Table 1 reporting the forecasting performance of monitoring forecasts in terms of out-of-sample R^2 (%). Traditional robust monitoring forecasts are based on discounted mean square forecast error (DMSFE) from [Stock and Watson \(2004\)](#) and logistic regressions. Each column corresponds to a different sample split year from 1947 to 2007. All evaluation period ends in December 2017.

Forecasts	First year in evaluation sample						
	1947	1957	1967	1977	1987	1997	2007
Proposed forecast $f^{(a)}$	0.50	0.37	0.36	0.14	-0.09	-0.10	-0.24
Robust monitoring machine: $f^{(m)}$	0.57	0.55	0.52	0.34	0.35	0.32	0.18
Shrinkage: $(f^{(a)} + f^{(b)})/2$	0.29	0.21	0.21	0.09	-0.02	-0.03	-0.12
Traditional robust monitoring forecasts							
DMSFE (60 months, $\delta = 1.0$)	0.40	0.30	0.39	0.24	0.04	0.01	-0.10
DMSFE (60 months, $\delta = 0.5$)	0.45	0.42	0.38	0.19	0.10	0.14	0.09
Logistic regression (feature engineering)	0.21	0.11	0.06	-0.05	-0.36	-0.48	-0.46
Logistic regression (no feature engineering)	0.37	0.23	0.25	0.15	-0.08	-0.30	-0.60

Table 5: Certainty Equivalent Return Gains of Monitoring Forecasts

This table repeats Table 4 reporting the forecasting performance of monitoring forecasts, but in terms of certainty equivalent returns (% , annualized). Traditional robust monitoring forecasts are based on discounted mean square forecast error (DMSFE) from [Stock and Watson \(2004\)](#) and logistic regressions. Each column corresponds to a different sample split year from 1947 to 2007. All evaluation period ends in December 2017.

	First year in evaluation sample						
	1947	1957	1967	1977	1987	1997	2007
Proposed forecast $f^{(a)}$	0.90	0.77	0.81	0.33	0.10	0.36	0.40
Robust monitoring machine: $f^{(m)}$	1.05	1.08	1.13	0.81	0.92	1.22	0.84
Shrinkage: $(f^{(a)} + f^{(b)})/2$	0.51	0.44	0.46	0.21	0.10	0.23	0.18
Traditional robust monitoring forecasts							
DMSFE (60 months, $\delta = 1.0$)	0.69	0.59	0.76	0.45	0.26	0.30	0.54
DMSFE (60 months, $\delta = 0.5$)	0.91	0.94	0.90	0.51	0.48	0.63	0.67
Logistic regression (feature engineering)	0.54	0.35	0.44	0.30	0.06	-0.33	-0.61
Logistic regression (no feature engineering)	0.46	0.36	0.30	0.13	-0.17	0.01	0.14

Appendix A A List of Individual Predictor Variables

We list the names and brief descriptions of 14 predictor variables we use as follows:

1. Dividend-price ratio (dp): log of a 12-month moving sum of dividends paid on the S&P 500 Index minus the log of stock prices (S&P 500 Index)
2. Dividend yield (dy): log of a 12-month moving sum of dividends minus the log of lagged stock prices.
3. Earning-price ratio (ep): log of a 12-month moving sum of earnings on the S&P 500 Index minus the log of stock prices.
4. Dividendpayout ratio (de): log of a 12-month moving sum of dividends minus the log of a 12-month moving sum of earnings.
5. Equity risk premium volatility (rvol): based on a 12-month moving standard deviation estimator
6. Book-to-market ratio (bm): book-to-market value ratio for the Dow Jones Industrial Average
7. Net equity expansion (ntis): ratio of a 12-month moving sum of net equity issues by NYSE-listed stocks to the total end-of-year market capitalization of New York Stock Exchange (NYSE) stocks.
8. Treasury bill rate (tbl): interest rate on a three-month Treasury bill (secondary market).
9. Long-term yield (lty): long-term government bond yield.
10. Long-term return (ltr): return on long-term government bonds.
11. Term spread (tms): long-term yield minus the Treasury bill rate.
12. Default yield spread (dfy): difference between Moodys BAA- and AAA-rated corporate bond yields.
13. Default return spread (dfr): long-term corporate bond return minus the long-term government bond return.
14. Inflation (infl): calculated from the CPI for all urban consumers.

Appendix B Machine Learning Algorithms used in the Paper

1. Random Forest This is an bagging method that combines forecasts from many small decision trees. This algorithm introduces extra randomness when growing trees. Instead of trying to fit the whole sample, each small decision tree will only use a sub-sample of the dataset. Averaging the results from small decision trees can improve the predictive accuracy and control over-fitting. This brings greater diversity, which trades a higher bias for a lower variance and yields an overall better model. This algorithm repeats the process for multiple times while producing one decision tree for each time. At the end, it combines multiple decision trees and averages their forecasts. We use default loss function Gini impurity of RandomForest in scikit-learn package.

2. Extremely Randomized Trees Instead of using sub-sample of the dataset and searching for the best possible threshold for each feature when splitting a node, this algorithm uses the full sample and random thresholds for each feature. This trades more bias for a lower variance and usually will be faster to train than regular random forest since finding the best possible threshold for each feature at every node is very time-consuming. We use default loss function Gini impurity of ExtraTrees in scikit-learn package.

3. Gradient Boosting This algorithm works by sequentially adding a new decision tree to an ensemble of previous trees with each new one trying to correct the forecasting errors from its predecessor. It fit the new predictor to the residual errors made by the previous predictor. Shallow trees on their own are “weak learners” with weak predictive power. The theory behind boosting suggests that many weak learners may, as an ensemble, comprise a single “strong learner” with greater stability than a single complex tree. We use default loss function friedman-mse of GradientBoosting in scikit-learn package.